

CPS I — Exercise Sheet 1

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1. Propositional Logic

Atomic Propositions

- a Alice joins group 1
- b Bob joins group 1
- c Claire joins group 1

Assign the value true if the corresponding person joins group 1, and false else

Given Conditions

1. If Alice joins group 1, the tutor refuses to accept Bob.

Formula: $a \rightarrow \neg b$

2. At least one of Bob and Claire cannot go to group 1.

Formula: $\neg(b \wedge c)$

3. Claire hates Alice and doesn't want to be in the same group.

Formula: $a \leftrightarrow \neg c$

4. Alice wants to submit solutions with either Bob or Claire.

Formula: $a \rightarrow ((a \leftrightarrow b) \vee (a \leftrightarrow c))$

Truth Table

a	b	c	1	2	3	4	Overall Logic
T	T	T	F	F	F	T	F
T	T	F	F	T	T	T	F
T	F	T	T	T	F	T	F
T	F	F	T	T	T	F	F

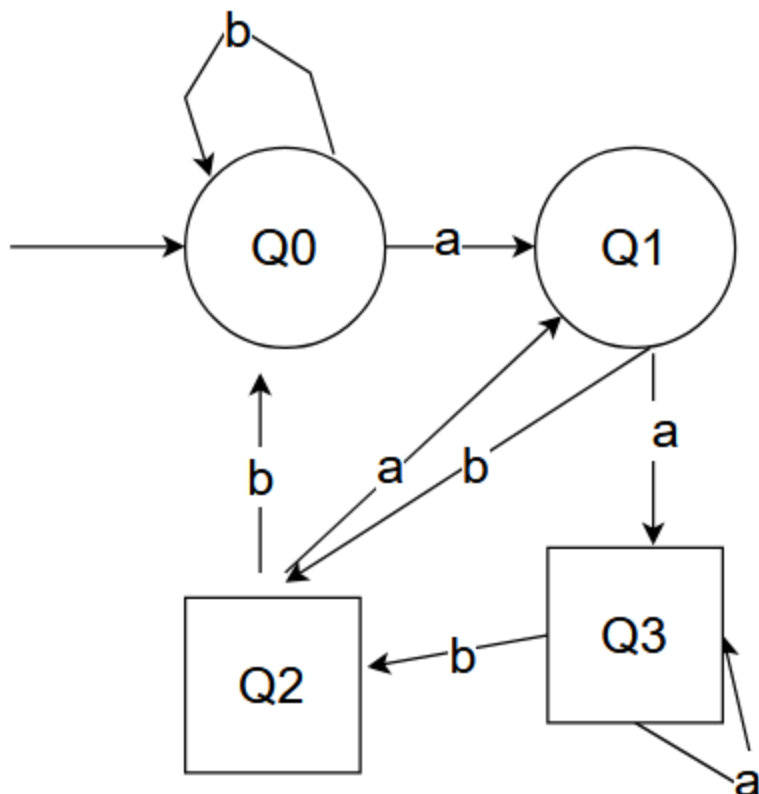
a	b	c	1	2	3	4	Overall Logic
F	T	T	T	F	T	F	F
F	T	F	T	T	T	T	F
F	F	T	T	T	T	T	T
F	F	F	T	T	F	T	F

Conclusion: $a = \text{False}, b = \text{False}, c = \text{True}$

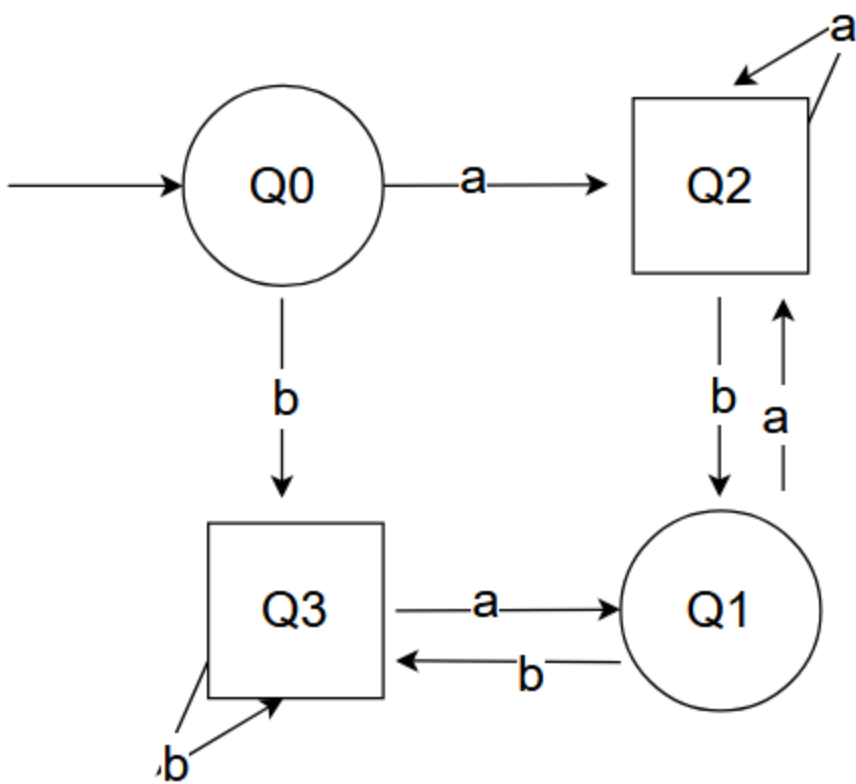
2. Finite Automata

- (a) $L_1 = \{x_0x_1 \dots x_n \mid n \in \mathbb{N}, n \geq 1, (\forall i \leq n, x_i \in \{a, b\}), x_{n-1} = a\}$
- (b) $L_2 = \{x_0x_1 \dots x_n \mid n \in \mathbb{N}_0, (\forall i \leq n, x_i \in \{a, b\}), x_0 = x_n\}$
-

L1:



L2



• L1

- $Q: \{Q_0, Q_1, Q_2, Q_3\}$
- $\Sigma: \{a, b\}$
- $\delta: \begin{cases} (Q_0, a, Q_1), (Q_0, b, Q_0) \\ (Q_1, a, Q_3), (Q_1, b, Q_2) \\ (Q_3, a, Q_3), (Q_3, b, Q_2) \\ (Q_2, a, Q_1), (Q_2, b, Q_0) \end{cases}$
- $Q_{\text{init}}: \{Q_0\}$
- $F: \{Q_2, Q_3\}$

• L2

- $Q: \{Q_0, Q_1, Q_2, Q_3\}$
- $\Sigma: \{a, b\}$
- $\delta: \begin{cases} (Q_0, a, Q_2), (Q_0, b, Q_3) \\ (Q_2, a, Q_2), (Q_2, b, Q_1) \\ (Q_1, a, Q_2), (Q_1, b, Q_3) \\ (Q_3, a, Q_1), (Q_3, b, Q_3) \end{cases}$
- $Q_{\text{init}}: \{Q_0\}$
- $F: \{Q_2, Q_3\}$